

Level - 2

■ Ionic Bond

- The ionic bonds X^+Y^- are formed when :
 (I) electron affinity of Y is high
 (II) ionization energy of X is low
 (III) lattice energy of XY is high
 (IV) lattice energy of XY is low
 Choose the correct code :
 (a) I and II (b) I and III (c) I, II and III (d) All
- In the Born-Haber cycle for the formation of solid common salt (NaCl), the largest contribution comes from:
 (a) the low ionization potential of Na
 (b) the high electron affinity of Cl
 (c) the low ΔH_{vap} of Na(s)
 (d) the lattice energy
- Incorrect statement is :
 (a) $\text{MgO} > \text{AlF}_3 > \text{MgF}_2$: Lattice energy
 (b) $\text{Li} > \text{Na} > \text{Al} > \text{Mg}$: Electron affinity
 (c) $\text{SF}_6 > \text{PF}_5 > \text{SiF}_4$: Lewis acidic character
 (d) $\text{SiCl}_4 > \text{SiBr}_4 > \text{SiI}_4$: Decreasing order of electronegativity of Si

■ Valence Bond Theory

- If two different non-axial d -orbitals having 'xx' nodal plane form π -bond by overlapping each other, then internuclear axis will be :
 (a) x (b) y
 (c) z (d) They don't form π -bond
- Assuming pure $2s$ and $2p$ orbitals of carbon are used in forming CH_4 molecule, which of the following statement is false?
 (a) Three C—H bonds will be at right angle
 (b) One C—H bond will be weaker than other three C—H bonds
 (c) The shape of molecule will be tetrahedral
 (d) The angle of C—H bond formed by s - s overlapping will be uncertain with respect to other three bonds.
- The strength of bonds formed by $2s$ - $2s$, $2p$ - $2p$ and $2p$ - $2s$ overlap has the order :
 (a) $s-s > p-p > p-s$ (b) $s-s > p-s > p-p$
 (c) $p-p > p-s > s-s$ (d) $p-p > s-s > p-s$
- Which of the following statements is incorrect for sigma and π -bonds formed between two carbon atoms?
 (a) Sigma-bond is stronger than a π -bond
 (b) Bond energies of sigma and π -bonds are of the order of 264 kJ/mol and 347 kJ/mol
 (c) Free rotation of surrounding atoms about a sigma-bond is allowed but not in case of a π -bond
 (d) Sigma-bond determines the direction between carbon atoms but a π -bond has no primary effect in this regard
- Assuming the bond direction to the z -axis, which of the overlapping of atomic orbitals of two atom (A) and (B) will result in bonding?
 (I) s -orbital of A and p_x -orbital of B (II) s -orbital of A and p_z orbital of B
 (III) p_y -orbital of A and p_z orbital of B (IV) s -orbital of both (A) and (B)
 (a) I and IV (b) I and II (c) III and IV (d) II and IV

9. Which of the following orbital can not form π as well as δ -Bond ?

(a) d_{xy}
(b) d_{z^2}
(c) $d_{x^2-y^2}$
(d) d_{yz}

VSEPR Theory

10. Which of the following statement is correct about I_3^+ and I_3^- molecular ions ?
(a) Number of lone pairs at central atoms are same in both molecular ions
(b) Hybridization of central atoms in both ions are same
(c) Both are polar species
(d) Both are planar species
11. In which of the following molecular shape d_{z^2} -orbital must not be involved in bonding ?
(a) Pentagonal planar (b) Trigonal planar (c) Linear (d) Square planar
12. The correct statement regarding SO_2 molecule is:
(a) two $p\pi - d\pi$ bonds
(b) molecule has 2 lone pair, 2 σ bonds and 2 π bonds
(c) two $p\pi - p\pi$ bonds
(d) one $p\pi - p\pi$ and one $p\pi - d\pi$ bond
13. A molecule XY_2 contains two σ , two π -bonds and one lone pair of electron in the valence shell of X. The arrangement of lone pair as well as bond pairs is :
(a) square pyramidal (b) linear (c) trigonal planar (d) unpredictable
14. In which of the following pairs, both the species have the same hybridisation ?
(I) SF_4 , XeF_4 (II) I_3^- , XeF_2 (III) ICl_4^+ , SiCl_4 (IV) ClO_3^- , PO_4^{3-}
(a) I, II (b) II, III (c) II, IV (d) I, II, III
15. Which of the following possess two lone pair of electrons on the central atom and square planar in shape ?
(I) SF_4 (II) XeO_4 (III) XeF_4 (IV) ICl_4^-
(a) I, III (b) II, IV (c) III, IV (d) All
16. Select pair of compounds in which both have different hybridization but have same molecular shape :
(a) BF_3 , BrF_3 (b) ICl_2^+ , BeCl_2 (c) BCl_3 , PCl_3 (d) PCl_3 , NCl_3
17. The species having no $p\pi - p\pi$ bond but has bond order equal to that of O_2 :
(a) ClO_3^- (b) PO_4^{3-} (c) SO_4^{2-} (d) XeO_3

Hybridisation

18. Choose the correct code of characteristics for the given order of hybrid orbitals of same atom,

$$sp < sp^2 < sp^3$$

(i) Electronegativity

(iii) Size

(a) (ii), (iii) and (iv)

(c) (ii) and (iv)

(ii) Bond angle between same hybrid orbitals

(iv) Energy level

(b) (iii), (iv)

(d) (i), (ii), (iii) and (iv)

19. Which is correct statement?

As the s-character of a hybrid orbital decreases

(I) The bond angle decreases

(III) The bond length increases

(II) The bond strength increases

(IV) Size of orbitals increases

- (a) (I), (III) and (IV)
(c) (I) and (II)

20. Which of the following is incorrectly match ?

Hybridisation

- (a) sp^3d
(b) sp^3d^3
(c) sp^3d^2
(d) sp^3

Geometry

- Trigonal bipyramidal
Pentagonal bipyramidal
Capped octahedral
Tetrahedral

- (b) (II), (III) and (IV)
(d) All are correct

Orbitals use

- $s + p_x + p_y + p_z + d_{z^2}$
 $s + p_x + p_y + p_z + d_{x^2-y^2} + d_{z^2} + d_{xy}$
 $s + p_x + p_y + p_z + d_{x^2-y^2} + d_{z^2}$
 $s + p_x + p_y + p_z$

Molecular Geometry/Shape

21. Species having maximum 'Cl—O' bond order is :
(a) ClO_3^- (b) ClO_3 (c) ClO_2 (d) ClO_2^-
22. Which of the following species contains minimum number of atoms in 'XY' plane?
(a) XeF_5^- (b) SF_6 (c) IF_7 (d) All
23. The molecule ML_x is planar with 7 pairs of electrons around M in the valence shell. The value of x is.
(a) 6 (b) 5 (c) 4 (d) 3
24. Choose the correct option for the following molecule in view of chemical bonding :
- $$\begin{array}{c} \text{Cl} \\ \diagup \\ \text{H} \end{array} \text{C} = \text{C} = \text{C} = \text{C} \begin{array}{c} \text{H} \\ \diagdown \\ \text{Cl} \end{array}$$
- (a) non-planar (b) $\mu \neq 0$ (c) both (a) and (b) (d) $\mu = 0$

Bond Parameters

25. Which of the following fact is directly explained by the statement 'oxygen is a smaller atom than sulphur'
(a) H_2O boils at a much higher temperature than H_2S
(b) H_2O undergoes intermolecular hydrogen bonding
(c) H_2O is liquid and H_2S is gas at room temperature
(d) S—H bond is longer than O—H bond
26. Which of the following compound has maximum "C — C" single bond length ?
(a) CH_2CHCCH (b) HCCCCH
(c) CH_3CHCH_2 (d) $\text{CH}_2\text{CHCHCH}_2$
27. Which of the following set contains species having same angle around the central atom?
(a) SF_4 , CH_4 , NH_3 (b) NF_3 , BCl_3 , NH_3
(c) BF_3 , NF_3 , AlCl_3 (d) BF_3 , BCl_3 , BBr_3
28. Which of the following compound has the smallest bond angle (X — A — X) in each series respectively
(A) OSF_2 (B) SbCl_3 (C) PI_3 (D) OSCl_2 (E) SbBr_3 (F) AsI_3 (G) OSBr_2 (H) SbI_3 (I) SbI_3
(a) OSF_2 , SbCl_3 and PI_3 (b) OSBr_2 , SbI_3 and PI_3
(c) OSF_2 , SbI_3 and PI_3 (d) OSF_2 , SbCl_3 and SbI_3

Molecular Forces

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29. The incorrect order of boiling point is :
 (a) $\text{H}_2\text{O} > \text{CH}_3\text{OH}$
 (c) $\text{H}_3\text{PO}_4 > \text{Me}_3\text{PO}_4$
 (b) $\text{CH}_3\text{NH}_2 < \text{NH}(\text{CH}_3)_2$
 (d) $\text{CH}_3\text{N}_3 > \text{HN}_3$
30. Iodine molecules are held in the solid lattice by
 (a) London forces
 (c) covalent bonds
 (b) dipole-dipole interactions
 (d) coulombic force
31. Carbon dioxide is gas, while SiO_2 is solid because :
 (a) CO_2 is a linear molecule, while SiO_2 is angular
 (b) van der Waals' forces are very strong in SiO_2
 (c) CO_2 is covalent, while SiO_2 is ionic
 (d) Si cannot form stable bonds with O, hence Si has to form a 3D lattice
32. LUMO [Lowest Unoccupied Molecular Orbital] of which of the following paramagnetic molecular species is not formed by s - p mixing:
 (a) O_2^+
 (b) N_2^{2+}
 (c) B_2^-
 (d) NO
33. In which of the following molecular species s -bond is having fractional bond order equal to 0.5?
 (a) N_2^-
 (b) O_2^-
 (c) NO
 (d) N_2^+
34. Select incorrect statement for a diatomic homonuclear +2 cation having molecular orbital configuration $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p)^2(\pi 2p)^4$
 (a) Ion is diamagnetic
 (b) It's neutral diatomic molecule is paramagnetic
 (c) s - p mixing occurs
 (d) It has higher bond order than respective neutral molecules
35. The O — O bond length in O_2 , $\text{O}_2[\text{AsF}_4]$ and KO_2 is:
 (a) $\text{O}_2[\text{AsF}_4] < \text{O}_2 < \text{KO}_2$
 (b) $\text{O}_2[\text{AsF}_4] < \text{KO}_2 < \text{O}_2$
 (c) $\text{O}_2 < \text{O}_2[\text{AsF}_4] < \text{KO}_2$
 (d) $\text{KO}_2 < \text{O}_2 < \text{O}_2[\text{AsF}_4]$
36. Which of the following transformation has identical LUMO on both side?
 (a) $\text{N}_2 \longrightarrow \text{N}_2^{2-}$
 (b) $\text{O}_2 \longrightarrow \text{O}_2^{2+}$
 (c) $\text{C}_2 \longrightarrow \text{C}_2^{2-}$
 (d) $\text{B}_2 \longrightarrow \text{B}_2^{2-}$

37.

List-I

- (P) $\text{S}_2(\text{g})$
 (Q) $\text{N}_2(\text{g})$
 (R) $\text{O}_2(\text{g})$
 (S) $\text{F}_2(\text{g})$

List-II (Properties)

- (1) Paramagnetic
 (2) Diamagnetic
 (3) Paramagnetic
 (4) Diamagnetic

Codes:

	P	Q	R	S
(a)	2	3	1	4
(b)	3	1	4	2
(c)	1	4	3	2
(d)	4	2	1	3

38.

List-I

- (P) B_2
 (Q) C_2
 (R) N_2
 (S) O_2

Codes:

	P	Q	R	S
(a)	1	3	4	2
(b)	4	1	2	3
(c)	3	2	1	4
(d)	4	2	1	3

List-II

- (1) Paramagnetic
- (2) Ionization energy is less than parent atom
- (3) Only π -bond is present
- (4) Bond order is equal to 3

► Level-2

1.	(c)	2.	(d)	3.	(b,c)	4.	(d)	5.	(c)	6.	(c)	7.	(b)	8.	(d)	9.	(b)	10.	(d)
11.	(b)	12.	(d)	13.	(c)	14.	(c)	15.	(c)	16.	(b)	17.	(d)	18.	(b)	19.	(a)	20.	(c)
21.	(b)	22.	(b)	23.	(b)	24.	(d)	25.	(d)	26.	(c)	27.	(d)	28.	(d)	29.	(d)	30.	(a)
31.	(d)	32.	(a)	33.	(d)	34.	(c)	35.	(a)	36.	(d)	37.	(c)	38.	(a)				